# Freight rate prediction

Validation approach and results

## Data and checks

48,000 labeled loads cover January-October 2025. There are 12,000 unlabeled November/December loads. Training has 300 missing weights and 374 missing market index values; numeric fields use median imputation. There are 0 duplicate predictor rows and 736 pickup/delivery pairs absent from training.

## Training and evaluation

Train through July to evaluate August, then through August to evaluate September. Select settings by mean development MAE and reserve October for testing. Imputers and encoders fit within each training split; IDs are excluded. The final models refit on all labeled rows.

## Model choice

Compare a median baseline, Ridge and three histogram gradient boosting settings. The selected full model is hgb_3; the six-field model is hgb_1. December uses a separate model because its inputs lack coordinates and market signals.

## October holdout

| Inputs | Model | MAE | RMSE | R2 | MAPE |
|---|---|---|---|---|---|
| Full | hgb_3 | $196.66 | $672.68 | 0.806 | 9.37% |
| Six fields | hgb_1 | $169.05 | $661.17 | 0.813 | 8.83% |

## MAE intervals

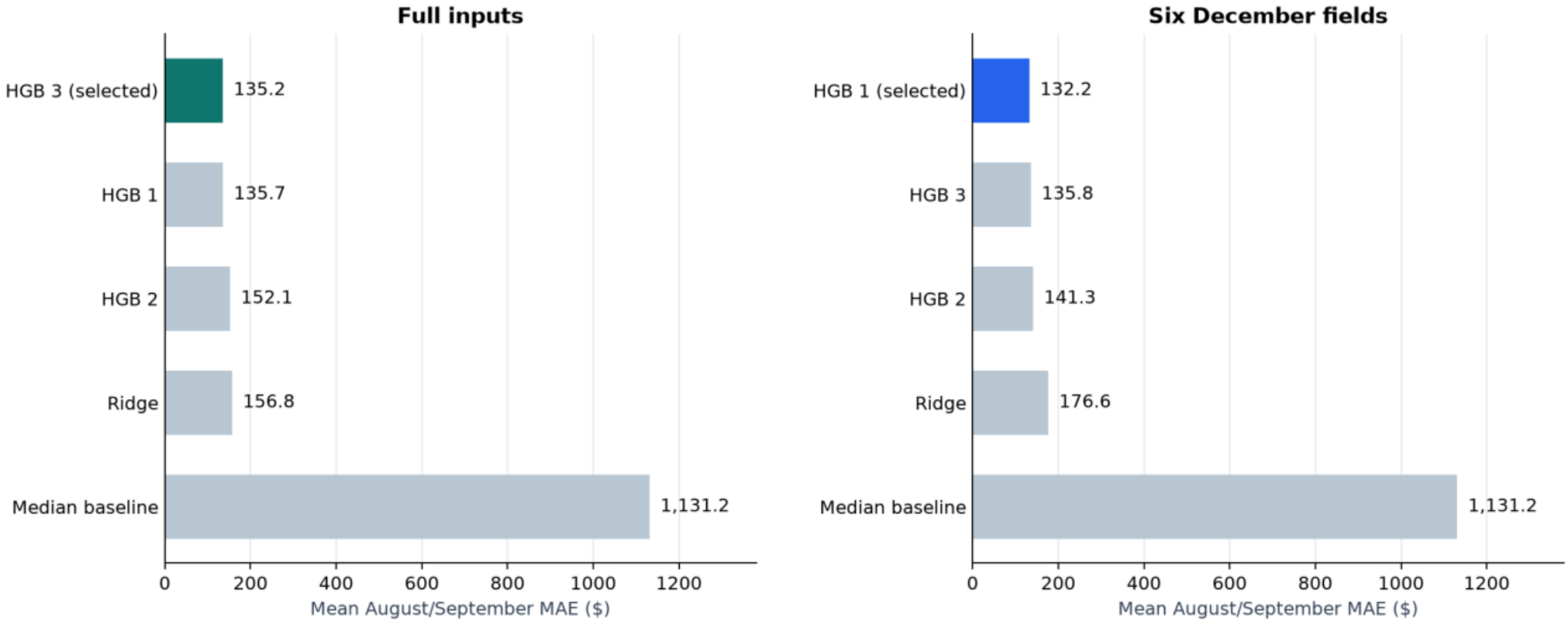
Resampling October days 1,000 times gives 95% intervals of $180.51-$212.46 for full inputs and $152.32-$186.38 for six fields. These describe variation across October days, not uncertainty for each future load.

## Limits

Future labels are unavailable. The two-month check overlaps September development data. Ten training months only partly cover annual seasonality, and trees have limited trend extrapolation. quote_signal is supplied for prediction; its availability before the quoted rate needs checking in production. The scorer checks file validity, not hidden-label accuracy.

# Model comparison

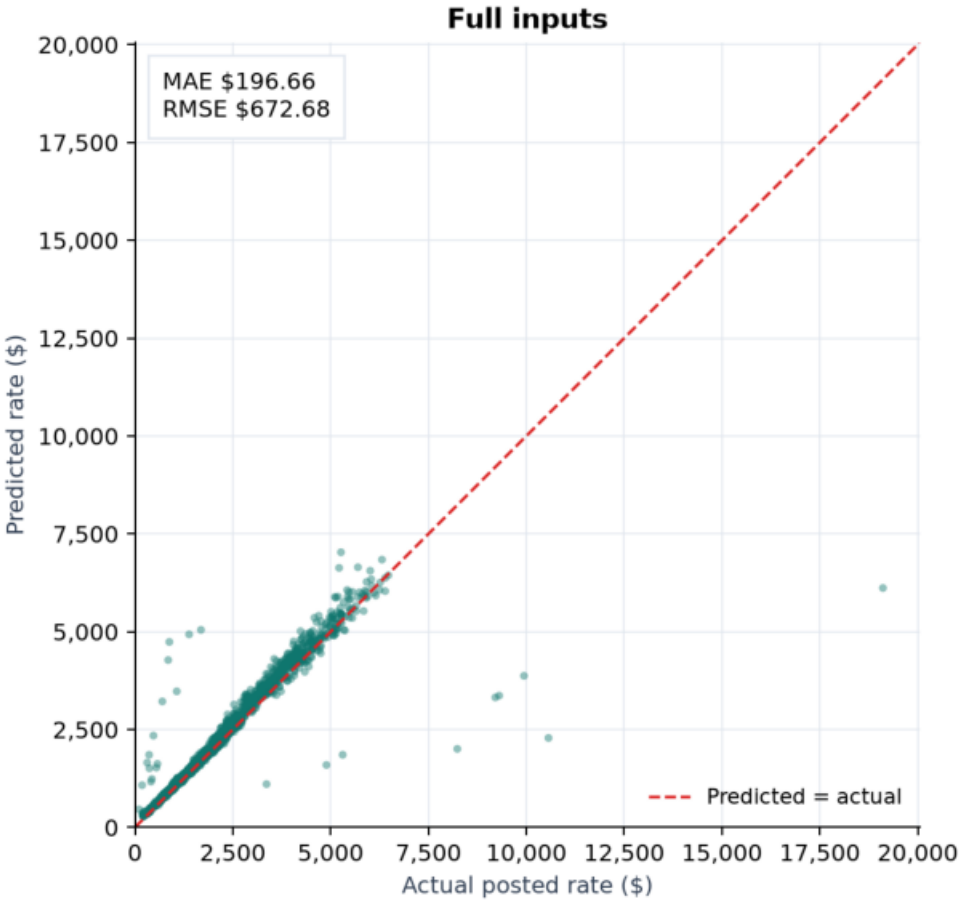
August and September development folds | lower MAE is better

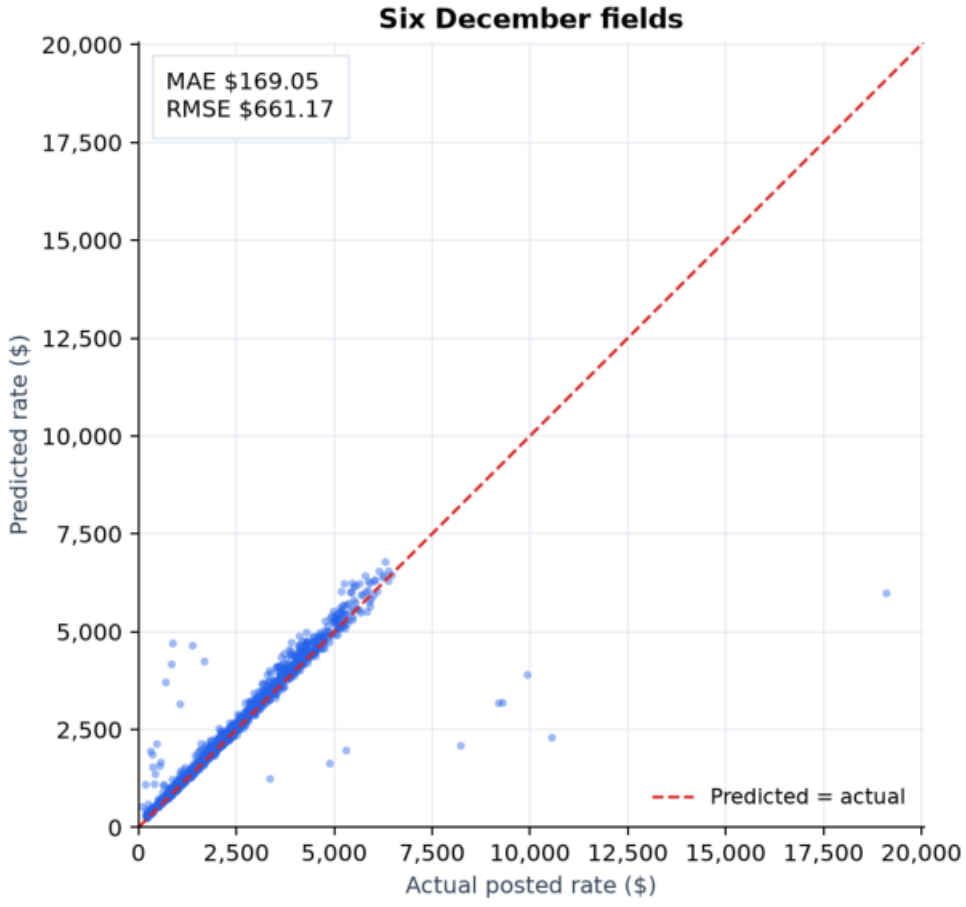



Each label shows mean development MAE in dollars. Colored bars mark the selected settings. Full-input HGB 1 and HGB 3 scores are close; October is excluded from this comparison.

# October prediction errors

4,853 held-out loads | 1,500 loads shown per model

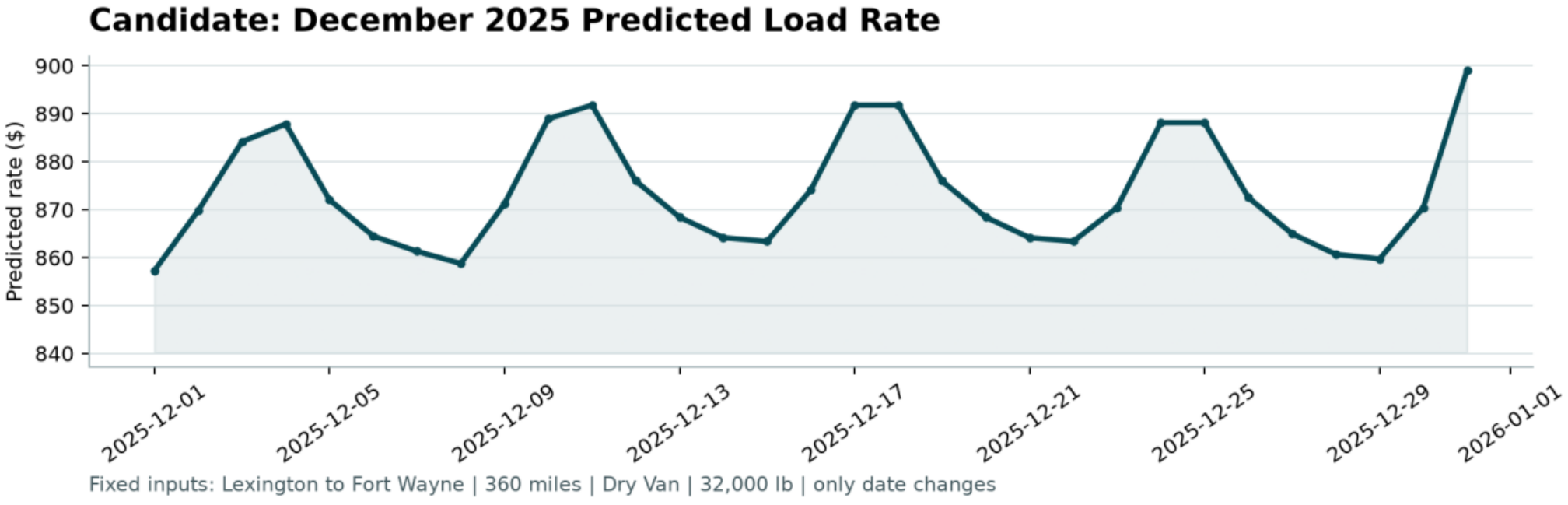



Both axes use the same scale. The dashed line marks an exact prediction. Some high-rate loads are underpredicted, which helps explain why RMSE exceeds MAE.

# December fixed-route forecast

Lexington to Fort Wayne | 360 miles | Dry Van | 32,000 lb




The original scorer chart uses a zoomed y-axis. Rates span $857.30-$899.06: a $41.76 range (4.8% of the mean). Only date changes. These are model forecasts; future labels are unavailable.